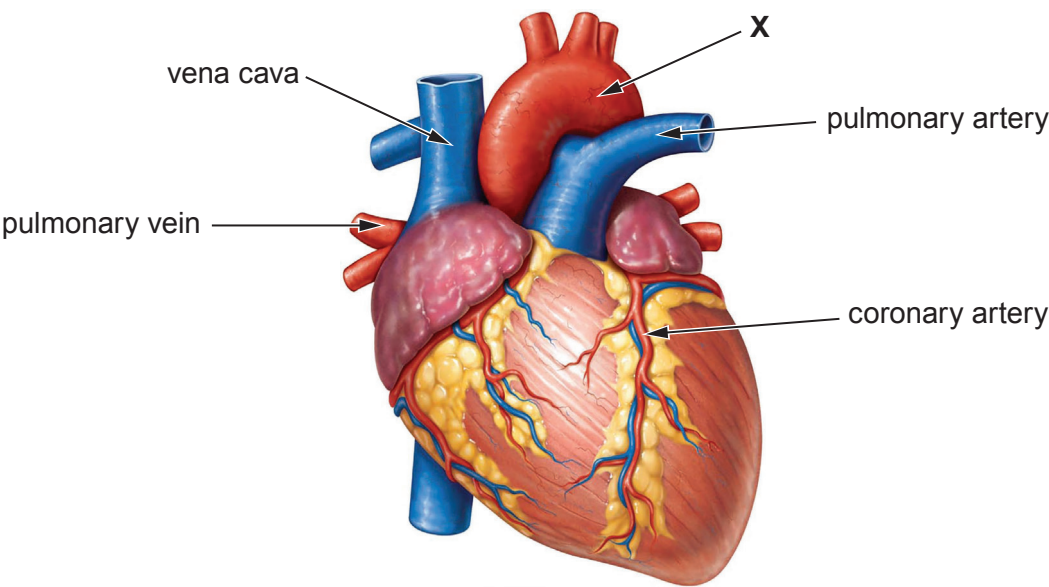


5. (a) The diagram shows the human heart.



(i) Blood vessel X is an artery carrying oxygenated blood from the heart at high pressure.
State the name of blood vessel X. [1]

.....

(ii) State the function of the coronary artery. [1]

.....

(b) Don investigated the effect of increasing intensity of exercise on the heart rate of three boys in his class, Dylan, Dewi and Dougie. His method is shown below.

- 1. He asked them to walk.
- 2. He counted their heart rate for one minute.
- 3. He repeated the method for jogging and then sprinting.

The results are shown in the table.

	Heart rate (beats/minute)		
	walking	jogging	sprinting
Dylan	74	84	96
Dewi	82	88	98
Dougie	80	89	98
Mean	79	87	97



Examiner
only

- (i) From the table:
- I. State the relationship between increasing intensity of exercise and heart rate. [1]

- II. Explain the importance of the relationship for respiring muscle tissue. [2]

- (ii) Suggest **two** ways Don could improve his experimental method in order to make a **fair** comparison between the three boys. [2]

- (iii) Suggest **one** way that Don could develop his investigation to make his results more representative of the whole population of Wales. [1]

8

